

Nirav Diwan

Ph.D. student in C.S., University of Illinois at Urbana-Champaign

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Education

Aug 2024 -	University of Illinois Urbana-Champaign (UIUC) Ph.D. in Computer Science Area: Security & Privacy, Machine Learning	Champaign, US
Aug 2022 May 2024	University of Illinois Urbana-Champaign (UIUC) Master of Science (M.S.) in Computer Science (CS) (thesis-track) Specialization: Machine Learning	Champaign, US
Aug 2017 Jun 2021	Indraprastha Institute of Information Technology, Delhi (IIITD) Bachelor in Technology (B.Tech) in Computer Science & Engineering (CSE) 🏆 Dean's Award for Academic Excellence	Delhi, India

Selected Publications

S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, J=Journal, *Equal Contribution

- [S.2] **PurpCode: Reasoning for Safer Code Generation** [🔗] [📄]
Jiawei Liu*, [Nirav Diwan*](#), Zhe Wang*, Haoyu Zhai, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, Muntasir Wahed, Yinlin Deng, Hadjer Benkraouda, Yuxiang Wei, Lingming Zhang, Ismini Lourentzou, Gang Wang
39th The Annual Conference on Neural Information Processing Systems [NeurIPS 2025]
🏆 **Won the 1st place in the Amazon Nova AI Challenge**
- [S.1] **MOCHA: Are Code Language Models Robust Against Multi-Turn Malicious Coding Prompts?** [🔗]
Muntasir Wahed, Xiaona Zhou, Kiet A. Nguyen, Tianjiao Yu, [Nirav Diwan](#), Gang Wang, Dilek Hakkani-Tür, Ismini Lourentzou
30th Empirical Methods in Natural Language Processing (Findings), 2025 [EMNLP 2025]
- [W.1] **Clear Preferences Leave Traces: Reference Model-Guided Sampling for Preference Learning** [🔗]
[Nirav Diwan](#), Tolga Ergen, Dongsub Shim, Honglak Lee
1st Workshop on Preparing Good Data for Generative AI Challenges and Approaches, 2025 [Good Data Workshop @ AAAI 2025]
- [C.4] **You Can't Judge a Binary by Its Header: Data-Code Separation for Non-Standard ARM Binaries using Pseudo Labels** [🔗]
Hadjer Benkraouda, [Nirav Diwan](#), Gang Wang
46th IEEE Symposium on Security and Privacy, 2025 [IEEE S&P 2025]
- [C.3] **It Doesn't Look Like Anything to Me: Using Diffusion Models to Subvert Visual Phishing Detectors** [🔗] [📄]
Qingying Hao, [Nirav Diwan](#), Ying Yuan, Mauro Conti, Giovanni Apruzzese, Gang Wang
33rd USENIX Security Symposium, 2024 [USENIX Security 2024]
- [C.2] **Weakening the Inner Strength: Spotting Core Collusive Users in YouTube Blackmarket Networks** [🔗] [📄]
Hridoy Sankar Dutta*, [Nirav Diwan*](#), Tanmoy Chakraborty
16th International AAAI Conference on Web and Social Media, 2022 [ICWSM 2022]
- [C.1] **Fingerprinting Fine-tuned Language Models in the Wild** [🔗] [📄]
[Nirav Diwan](#), Tanmoy Chakraborty, Zubair Shafiq
59th Annual Meeting of the Association for Computational Linguistics (Findings), 2021 [ACL 2021]
🏆 **Dean's Thesis Appreciation Award**
- [J.1] **RecipeDB: A Resource for Exploring Recipes** [🔗] [📄]
Devansh Batra*, [Nirav Diwan*](#), Utkarsh Upadhyay*, Jushaan Singh Kalra*, Tript Sharma*, Aman Kumar Sharma*, Dheeraj Khanna*, Jaspreet Singh Marwah*, Srilakshmi Kalathil*, Navjot Singh*, Rudraksh Tuwani*, Ganesh Bagler*
Database: The Journal of Biological Databases and Curation, Oxford University Press, 2020 [Database 2020]
🌐 **Press:** [Times of India](#), [The National](#), [Nature India](#)

Awards

Amazon Nova AI Challenge (Winner): Won 1st place as team co-lead in the [Amazon Nova AI Challenge](#) winning a cash prize of \$250K.

Amazon Nova AI Grant: Awarded grants worth \$250,000 + \$1.5 Million AWS Credits.

Catalyzing Advocacy in Science and Engineering (CASE) Workshop 2024: Only student selected to represent the CS department.

Dean's Thesis Appreciation Award, IIITD: For outstanding thesis research from Dean of Academic Affairs.

Dean's Award for Academic Excellence, IIITD: For exceptional academic performance for the academic years 2019-20.

IIT-JEE 2017: Ranked 1884/1.5 million (top 0.15%) in IIT-JEE Mains and 4186/200,000 (top 2%) in IIT-JEE Advanced.

Recent Industrial Experience

1. LG AI Research | Bilingual LLM Team [🌐]

Ann Arbor, US
- Research Intern | Area: Natural Language Processing (NLP), Safety and Alignment
- 2024
- Improved alignment of LLMs for single-turn and multi-turn settings. Results published in the **AAAI Good Data Workshop**.

➤ Developed a sampling method that outperformed alignment methods (DPO, ORPO, SimPO) for Coding, Math, and Reasoning.
2. Ema Inc. | Generative AI Team [🌐]

Remote, US
- Applied Science Intern | Area: Natural Language Processing (NLP)
- 2023
- Led the large-scale deployment of a conversational AI agent for translating natural language queries into SQL (Text2SQL).

➤ Built an eval pipeline and benchmark dataset of 3,000+ real-world Text2SQL examples for robust evaluation.

Selected Research Experience

1. University of Illinois Urbana-Champaign (UIUC) | Security & Privacy Group [🌐]

Champaign, US
- Graduate Researcher | Advisor: [Prof. Gang Wang](#) | Areas: Machine Learning (ML) and Security & Privacy (S&P)
- 2022 - Present
- Identifying and protecting against practical underperformance, attacks, and misuse of AI models.

➤ Published projects at **IEEE S&P** and **USENIX Security**.
2. IIIT Delhi | Laboratory for Computational Social Systems (LCS2) [🌐]

Delhi, India
- Undergraduate Researcher | Advisor(s): [Prof. Zubair Shafiq](#), [Prof. Tanmoy Chakraborty](#) | Areas: NLP, S&P and Graphs
- 2019 - 2021
- Identified malicious actors online using machine learning techniques.

➤ Published projects at **ACL** and **ICWSM**.

Teaching Experience

1. Introduction to Computer Science (CS124) | UIUC [🌐]

Champaign, US
- Teaching Assistant | Professor: [Prof. Geoffrey Challen](#)
- Fall 2022, Spring 2023, Fall 2023
- Served as Teaching Assistant for three consecutive semesters, engaging with 1,300+ students in each semester.

➤ Developed course materials, supervised quizzes, and held office hours.
2. Machine Learning Graduate (CS563) | IIIT Delhi

Delhi, India
- Teaching Assistant | Professor: [Prof. Tanmoy Chakraborty](#)
- Spring 2020
- Assisted teaching Graduate Machine Learning course to 180 students across undergraduate, master's, and doctoral levels.

➤ Conducted tutorials, resolved queries, graded quizzes, and developed the mid-semester examination.

Talks

- Can LLMs identify Security Vulnerabilities?

April 2025
- Advanced Computer Security, UIUC | [Slides](#)
- Jailbreaking of LLMs

October 2024
- LLM Post Pre-training, UIUC | [Slides](#)
- Generating Sequences by Learning to Self-Correct

May 2024
- Conversational AI, UIUC | [Slides](#)
- Promptly Racing Ahead: A Survey on Multi-task Prompt-Based Learning

May 2023
- Transfer Learning, UIUC | [Slides](#)
- Editing Models with Task Arithmetic

April 2023
- Transfer Learning, UIUC | [Slides](#)
- A pseudo-labelling approach for low-resource NLP Domain Adaptation

December 2022
- Trustworthy Machine Learning, UIUC | [Slides](#)
- Weight Poisoning Attack on Pre-trained Language Model

July 2021
- Security Machine Learning Seminar | [Slides](#)
- Practical No-box Adversarial Attacks against DNNs

June 2021
- Security Machine Learning Seminar | [Slides](#)

Academic Service

- Reviewer

ACM COMPASS'21, ICLR'22 (Student Reviewer)
- Volunteer

ACM FAccT'23, AAAI ICSWM'21, AAAI ICWSM'22, ACL'21